

Umer Irfan



Publications

Published

- Motion-Coupled Sensing: When the State Change Powers Its Own Sensing - IEEE MASS (Accepted)
- Kissan-Dost: Bridging the Last Mile in Smallholder Precision Agriculture with Conversational IoT - IEEE DCOSS (Accepted)

Submitted

- PARDA: On-Device Audio Privacy for Smart Glasses using Small Language Models - PerCom 2027 (Submitted)

Preprints (arXiv)

- LLM-Enhanced Wearables for Comprehensible Health Guidance in LMICs - arXiv:2602.08701
- WatchTower: A Transformer-Based Open-Set Network Anomaly Detection System for Edge Devices

Manuscripts in progress

- On-Chip Program Behavioral Adherence Monitoring for Runtime Hardware Trojan Detection via Neural Networks - (Targeting DAC 2027)
- Programmer-in-the-Middle: Attacks on Flash Programming Equipment for Stealthy Firmware Implantation

Other

- PixelPacker: Background-Aware Neural Image Compression for Wildlife Edge Systems (Paused)

About me

I'm a Senior at LUMS, majoring in Computer Science, with a minor in computer architecture.

Since 2014, I've immersed myself in programming, from game development to building my custom computer architectures on breadboards and FPGAs.

I thrive at the intersection of software and hardware.

I seek opportunities to innovate and create meaningful projects.

Education

Lahore University of Management Sciences

BS Computer Science

Graduating Class of 2027

SGPA: 3.707 (CS Courses)

CGPA: 3.491 (Including Humanities)

Pak-Turk Maarif International Schools and Colleges

A-levels (4A*)

Class of 2023

Pak-Turk Maarif International Schools and Colleges

O-levels (10 A*)

Class of 2021

Relevant Skills

C, C++, Python, Assembly, Verilog/SystemVerilog, AVR, ESP32, STM32, Raspberry Pi, FreeRTOS, RISC-V, MIPS, FPGA Design, RTL, Linux Kernel Development, Device Drivers, QEMU, Firmware Analysis, Binary Rewriting, Side-Channel Analysis, PCB Design, Digital Logic, SPI, I²C, UART, PyTorch, TensorFlow Lite Micro, TVM, Model Quantization



Past Teaching Experience

Teaching Assistant – ENGG100 (Engineering Design Fundamentals) | Dr. Zubair Khalid | 2024

Introduced 30 freshmen to core engineering skills through hands-on tutorials on Arduino, MATLAB, and Proteus; developed lab quizzes and grading rubrics for circuit simulation and design assignments.

Teaching Assistant – CS210 (Data Structures) | CS Core | Dr. Maryam Abdul Ghafoor | 2025

Collaborated with a 10-member course staff to design complex programming assignments and build automated grading systems. Delivered tutorials on core data structure concepts, evaluated quizzes, and adjudicated grade contestations.

Teaching Assistant – EE320 (Computer Organization & Assembly Language) | Grad Elective | Dr. Nadeem Khan | 2026

Sole TA. Revamped the course structure following David Patterson and John Hennessy's book. Authored 6 quizzes and 7 homework sets, and handled all grading, delivered tutorials and revision classes, invigilation, lab design, and evaluations.

Teaching Assistant – AI624 (AI on Edge Devices) | Grad Elective | Dr. Zubair Khalid | 2026

Authored a list of 8 course projects, class quizzes, and programming assignments on Knowledge Distillation and LLM Inference on low power devices, for a cohort of 82 graduate students as one of 4 course TAs.

Teaching Assistant – CS370 (Operating Systems) | CS Core | Dr. Naveed Anwar Bhatti | 2026

Designed assignments building a custom x86_64 kernel from BIOS handoff through 64-bit protected mode, including user-space shell implementation. Collaborated with an 8-member course staff to support a cohort of 295 students.

Course Designer - Diploma In Digital Health | LUMS Learning Institute (LLI) | 2026

Assisted with course design and delivered classes to a cohort of 90+ senior medical professionals for integrating Internet of Medical Things (IOMT) into their medical workflows.

Research Teams, Groups and Labs

Research Assistant - Embedded AI Research Group (SYSNET Lab) 2023 - 2026 | Dr. Naveed Anwar Bhatti

- Investigated transiently powered computing, edge AI, embedded malware injection and neural video compression.
- Developed hardware and software systems to extract insights into human behavior and serve as platforms for human interaction studies.
- Developed and evaluated embedded sensing systems for real-world deployments.
- Overlooked several Senior Year projects assisting as a hardware specialist.
- Contributed to peer-reviewed publications at IEEE MASS and DCOSS-IoT.

Research Assistant - Computing Hardware for AI & Security (CHASE) Lab 2025 - 2026 | Dr. Ali Siddiqi

- Developed FPGA datapaths for hardware-accelerated ML inference.
- Characterized power side-channel leakage in embedded computing platforms.
- Investigated firmware implantation and runtime hardware Trojan detection.

Projects In Progress

On-Chip Program Behavioral Adherence Monitoring for Runtime Hardware Trojan Detection via Neural Networks

- Investigating runtime detection of dormant hardware Trojans inserted during chip design and fabrication that pass through traditional validation steps.
- Developing an anomaly detection pipeline using transaction-level traces of on-chip program behavior
- Investigating the use of stateful 1D convolutional neural networks to identify deviations from expected program behavior.
- Developing a taxonomy of computational and application-based workloads to evaluate detection across different classes of programs and embedded system operations.
- Constructing paired datasets of expected and Trojan-affected execution traces for individual programs to support supervised classification.
- Investigating classifier generalization across different workloads, chip configurations and sizes, and hardware Trojan implementations.
- Investigating the effects of cached vs uncached implementations on observed transaction patterns and classifier performance.
- Reviewing runtime hardware Trojan defenses based on hardware performance counters, power and EM side channels, communication-channel statistics, control-flow attestation, and software instrumentation. Demonstrating a qualitative and quantitative comparison against our technique.
- Evaluating silicon overhead and timing characteristics of the final implemented module.
- Targeting an FPGA-based monitoring and classification architecture, with evaluation planned for detection accuracy, latency, memory requirements, and hardware resource utilization.

Programmer-in-the-Middle (PITM) Attacks on Flash Programming Equipment for Stealthy Firmware Implantation

- Investigating firmware supply-chain attacks through compromised flash programming equipment.
- Developed a threat model where the programmer controls both firmware writes and post-programming readback, allowing modified firmware to be concealed from host-side verification.
- Demonstrated PITM on USBasp, intercepting and modifying arbitrary programs to redirect execution to an injected firmware stub on AVR microcontrollers.
- Investigating in-place binary rewriting through pattern-based scanning and trampoline insertion, avoiding full control-flow graph recovery and intermediate representation reconstruction.
- Reverse-engineered ST-Link V2 firmware, identifying programming routines, flash algorithm pointers, and memory regions suitable for code injection.
- Developed custom ST-Link firmware files accepted by STM32CubeProgrammer, with compatibility tested across three target boards.
- Investigating integrity-check evasion through identification and modification of CRC comparison and conditional branch instructions in incoming firmware.
- Extending the attack across AVR, STM32, and Texas Instruments MSP430 programming architectures to examine applicability across different vendor toolchains.
- Preparing an evaluation of attack success, verification evasion, memory overhead, programming latency, and detectability through power and timing analysis.

Events & Honours

Winner - Pakistan Semiconductor Summit (PSS) Hackathon

2026

- Identified and fixed RTL bugs to ensure RISC-V specification compliance using SAIL.
- Developed verification tests and debugged processor implementations using Verilator
- Awarded 1st position with a team of 3 random students

Teknofest - Istanbul, Turkey

2020

- Authorized personnel detection project
- Interfaced Microsoft kinect with jettson nano.
- TensorFlow used to match faces with a database of authorised personnel.
- Raspberry Pi used to handle navigation and obstacle avoidance workloads.
- Interfaced the Raspberry Pi with the Nvidia Jetson Nano over UART.
- Designed power delivery circuitry with safety functionality and redundancy in mind.

Intel ISEF - Lahore, Pakistan

2019

- Constructed a SONAR based helmet to warn the visually impaired about approaching obstacles.
- Implemented temporal tracking of surrounding obstacles to form a rudimentary map.
- Utilised vibratory actuators to guide the user.
- Won the best project award at a provincial and national level.

IYPT - Beijing, China

2018

- Represented the team Pakistan in the International Young Physicists Tournament.
- Constructed scientific apparatus to enable time sensitive experiments.
- Researched novel physics phenomenon.
- Collaborated with national academic institutions (COMSATS, Fatimah Jinnah) to carry out physics research.

Hardware, Embedded Systems & Computer Architecture

Hand Made 8-Bit RISC CPU from Scratch (RIZZ-8)

2025

- Designed and built a custom 8-bit Harvard-architecture CPU with accumulator design, 16-bit address space, and 4 general-purpose registers.
- Exclusively used 74-series logic ICs, prototyped on breadboards, and fabricated 8+ custom dual-layer PCBs through JLCPCB with proper length matching and signal integrity practices.
- Developed a custom Turing-complete ISA supporting shifts, carries, signed and unsigned arithmetic, and boolean logic operations.
- Wrote a Python assembler to generate machine code.
- Demonstrated the pipeline end-to-end by writing ASM, flashing the machine code to EEPROMS, and running it in hardware.
- Learned about tri-state bus arbitration, synchronous timing design, and fundamental architecture principles.

Raspberry Pi Kernel Module

2023

- Developed a kernel driver that builds upon linux i2c subsystem.
- Created a device file that can be written to or read from to modify i2c devices' configuration.
- Minimised latency by optimizing transactions
- 1700 samples per second rate achieved
- Learned about Linux I2c subsystem
- Implemented checks in kernel space to ensure kernel panics don't occur while returning buffers to userspace

ATtiny85 Digital Level

2021 - 2022

- Interfaced MPU6050 with ATtiny85.
- Calculated angle using the accelerometer, used gyroscope to compensate for rotations.
- Created a watchdog timer based interrupt system to wake the mcu from sleep state.
- Custom PCB design using EasyEDA.
- Reached sub 0.0006 watt power consumption (20 microwatts in sleep state).
- Achieved and tested 30 day battery life using CR2025.
- Applied redundancy checks to ensure system performs reliably after various program state changes.
- Experienced working on embedded systems in a power constrained environment.

ESP-Now based Soil Monitoring Mesh Network

2024

- Developed a datalogger deployed in farms in Kasur to obtain PH, NPK concentrations, soil temperature, and moisture levels.
- Built a mesh network to allow multiple nodes to relay information to a root node.
- Uploaded the data to a MongoDB database.
- Ran time series analysis and forecasts, recommending actionable insights to the farmers.
- Created an LLM-powered pipeline, interfacing with the farmers via the WhatsApp API to allow farmers to provide conversational feedback on their crop.
- Conducted pre- and post-surveys to establish qualitative improvements in the fields across a time frame of 6 months.
- Research findings submitted to journals.

CARDY: A from Scratch ESP8266-Based companion for use in TechWars PSIFI '23 Hackathon

2023

- Developed an interactive challenge platform leveraging an ESP8266 microcontroller, implementing a CPP ESP8266WebServer.
- Used LittleFS (FAT-similar clone) to host and serve gzipped web assets (HTML, CSS, JS) and challenge files.
- Optimized web assets via Python scripts for inlining CSS/JS and Gzip compression, reducing file sizes and latency to serve resources from the onboard 2MB memory.
- Designed and implemented state-based HTTP endpoints to serve dynamic content based on users' progression, allowing users to resume the multi-stage technical challenges.
- Developed hardware interaction logic, including software-debounced button inputs, buzzers, and an LDR-based sensor for frequency detection (challenge input)
- Integrated and controlled a MAX7219 LED matrix display for scrolling text, challenge outputs (Morse code, binary patterns), and visual feedback.
- Automated the creation and manipulation of challenge assets using Python helper scripts and FFMPEG.
- Created a custom printed circuit board (PCB) manufactured from JLCPCB service (China) and procured parts via LUMS's procurement team from local vendors at Hall Road.
- Created a team of engineers to hand assemble 50 units of the CARDY device.
- Ran QA testing on the hardware units and utilised beta-testers to black-box test the hardware device.
- Utilised the WIFI enabled device to allow 50 teams from all over Pakistan to compete, while receiving realtime telemetry of every team's progress.
- Successfully conducted a hardware and software hackathon.

Machine Learning & Tiny ML

ESP32 and Tensorflow Lite Knock Detector 2022 - 2023

- Utilised Z axis of accelerometer to track knocking sound on a surface.
- Used 500 kilobit per second I2c mode to sample accelerometer at fastest possible rates.
- Created FreeRTOS tasks and utilised dual core functionality.
- Used core 1 to sample and fill circular buffer.
- Utilised Core 2 to run AI model using tensorflow micro library.
- Optimised to run model at 2.5 Hertz (faster than real time).
- Self trained AI model with pruning, 16 bit integer quantization and data augmentation.
- Experienced creating RTOS based projects on a memory and computationally constrained platform.

High Data Rate Fitness Band with Actionable feedback 2024

- Developed an indigenous high data rate ATmega328p-powered health band.
- Applied weighted quantization, density mapping, Descreeete Cosine Transformations (DCT) to maximise data in SRAM.
- Implemented data structures such as circular buffers, sampling at varying rates.
- Used literature to select minimum sample rates to retain 95 % accuracy when evaluating metrics.
- Created an LLM-powered pipeline, interfacing with users via the WhatsApp API to enable conversational feedback on the user's health.
- Conducted pre- and post-surveys to establish qualitative improvements in the behaviour towards smart bands.
- Research findings submitted to journals.

Soil Salinity and Moisture Prediction from LANDSAT-8 2023

- Processed Landsat 8 satellite images spanning 2013-2023, extracting infrared bands 6 and 7 to calculate Soil Salinity Index (SSI) using spectral analysis.
- Implemented ORB feature detection-based image alignment to ensure temporal consistency across multi-year satellite imagery datasets.
- ConvLSTM2D architecture was used for spatiotemporal pattern recognition in time-series satellite data.
- Created frame sequences to enable next-frame prediction of soil conditions in a patch, implementing early stopping and learning rate reduction callbacks for optimal training.
- Forecasted soil salinity and moisture trends for agricultural planning and environmental monitoring applications.

AI Voice Agent for Call Center Automation 2023-2024

- Developed an end-to-end application with NodeJS backend, streaming audio data over websockets to a proprietary Python backend on a DigitalOcean droplet.
- Real-time voice activity detection using WebRTC VAD for speech boundary detection.
- Integrated OpenAI Whisper API for transcription and GPT-4 models for conversational responses
- Created a PyAudio-based audio streaming pipeline with sampling rate matching.
- Implemented session handling using JSON.
- Built a MySQL database integration for user management and call logging.
- Engineered a prompt system supporting personas, such as sales reps, for demonstrations
- Achieved sub-2-second response latency from speech stream input to audio output through efficient API integration and audio buffering.

Tensorflow-based Voice Gender Classification 2022

- Developed audio processing pipeline using Librosa for resampling (to 8 kHz) and segmenting training files into 2-second clips.
- Generated time-frequency features by computing Spectrograms using Short-Time Fourier Transform (STFT).
- Applied audio data augmentation techniques (Pitch Shift, Noise, Room Simulation) using Audiomentations to create a large training set.
- Used Transfer Learning with a trained EfficientNetB0 backbone.
- Utilized TensorFlow's tf.data API for building an efficient input pipeline to reduce memory bottlenecks.
- Successfully trained the model to classify audio events based on learned patterns in spectrogram representations.

WatchTower: Transformer-Based Network Anomaly Detection for Edge Devices 2025

- Developed a transformer-based network anomaly detection system for Raspberry Pi 5, achieving real-time anomaly detection on resource-constrained edge hardware.
- Engineered a feature extraction pipeline processing 28 packet-level IP features including payload entropy, inter-arrival times, and protocol-specific flags from CIC-IDS2017 dataset and our own WireShark captures.
- Implemented PROSER open-set recognition via manifold mixup in latent space to detect threats not present in training data.
- Integrated FedProx federated learning to prevent catastrophic forgetting across distributed edge networks with non IID traffic patterns.
- Introduced a retraining schedule based on overall confidence scores to ensure the true class doesn't drift as network patterns change.
- Designed a novel memory token mechanism with gated attention updates to preserve long-term context across sliding windows of network packets.
- Applied mixed precision quantization and TVM auto-tuning targeting ARM Cortex-A76 to achieve a 3.82ms inference latency while maintaining 93.66% accuracy.
- Outperformed LSTM-Autoencoder baseline by 19% accuracy and 10% F1-score, demonstrating superior long-term temporal dependency modeling for network traffic.
- Validated on CIC-IDS2017 dataset and live DoS attacks using LOIC tool, achieving instantaneous threat detection with clear latent space separation between benign and malicious traffic.
- Achieved 2.6× speedup through hardware-aware compiler optimization and 67.5% model compression via INT8 quantization, enabling 260+ packets/sec throughput on edge hardware.

Software Engineering and Cloud Development

Makers Lab Management Dashboard 2024

- Developed a full-stack web application using React.js frontend and Node.js/Express backend for National Incubation Center's Makers Lab's inventory and task tracking system.
- Used PostgreSQL database for CRUD operations for employees, machines, tasks, and activity logs management.
- Created a role-based authentication system with cookie-based session management.
- Built RESTful APIs with 15+ endpoints.
- Designed React components for dynamic navigation for a multi-page dashboard interface.
- Integrated parameterized SQL queries with parsing to prevent SQL injection.
- Deployed several docker containerized application supporting multi-user concurrent access.

Chat Application with Go and MongoDB 2023

- Developed a distributed chat server using Go.
- Used HTTP endpoints for message posting and retrieval.
- Implemented MongoDB Atlas integration using certificate authentication.
- Created RESTful APIs for chat functionality.
- Built a command-line chat client with automatic server discovery, connection retry logic, and message polling.
- Implemented message sorting by timestamp with 100 most recent messages loaded.
- Achieved real-time messaging capabilities with immediate message posting and retrieval through HTTP-based communication protocol.

Redis Cache Performance Analysis Experimentation 2022

- Built a TypeScript-based Express.js server with PostgreSQL integration using ElephantSQL cloud db for user authentication queries and dummy data.
- Implemented a Redis caching layer with configurable TTL to optimize database query performance and reduce latency for repeated username lookups.
- Developed scripts for query performance measurement system tracking execution times in milliseconds with cache hit/miss analytics.
- Optimized and achieved sub-10ms query response times through Redis caching (on a local server) implementation, reducing database load by 80% for repeated queries.

Twitter Image Scraping and Classification System (POKI-API)

2022

- Built an automated Twitter scraping pipeline using Python to collect and process images from social media accounts.
- Performed coordinate extraction and automated cropping of faces.
- Used open-source CNN-based emotion detection models to classify and sort images based on emotions.
- Created a SHA256-based redundant image removal system.
- Built Express.js API endpoints for image serving, and automated directory management.
- Developed an automated workflow processing thousands of images with batch processing and error handling capabilities
- The program was run on a local server every 10 hours using cron jobs.

C++ Console ASCII Raycasting Engine 2023

- Developed a 3D raycasting engine from scratch in C++ for rendering a 2D grid map in a console environment.
- Implemented player movement within the 2D map using Win32 API (GetAsyncKeyState, Sleep) for real-time keyboard input detection and game loop timing control.
- Designed ray intersection logic to determine the distance from the player to the nearest wall.
- Implemented distance-based shading and brightness by mapping wall distances to a custom character set and calculating grayscale values, creating a visual depth.
- Projected rendered frames onto a 2D console screen buffer.
- Created an interactive Doom-style raycasted console game.

JPEG Image Decoder from Scratch in C++ 2023

- Developed a JPEG image decoder compliant with ISO/IEC 10918, decoding compressed image data into RGB color space from YCbCr values using C++.
- Focused on mastering the C++ STL.
- Efficiently processed compressed data using bit manipulation techniques, bitwise shifting, and bit masking.
- Implemented canonical Huffman encoding.
- Applied quantization tables during the inverse discrete cosine transform (IDCT).
- Demonstrated the use of pointer-based data structures to manage image-related information.
- Ensured robustness and stability through error handling and validation checks during decoding..

Community & Leadership Experience

Society for the Promotion and Development of Engineering and Sciences (SPADES) - TechWars Team Member 2023-2024

- Developed and hosted the CARDY hackathon along with a selected group of 8 members.
- Distributed and collaborated in managing workload, challenging design, and procurement.
- Generated, verified, sent, and inspected procurement orders, and managed interactions with different vendors.
- Collaborated with the Logistics department of the society to allow efficient handling of logistics.
- Catered to 250 participants over 5 days, managing event outlook and progress.

Society for the Promotion and Development of Engineering and Sciences (SPADES) - Convener LUMS Science Symposium (LSS) 2024

- Led a week-long 550-participant research-oriented program.
- Oversaw a team of 150 members spanning seven departments such as Registrations, Marketing, Media, and HR, keeping a check and balance on progression.
- Interfaced with the LUMS Office of Student Affairs (OSA) and other departments in operating the event within university guidelines, with subsequent approvals.
- Arranged sponsorships in kind and cash, amounting to 200k rupees.
- Generated a net profit of 800k rupees for the society.

Society for the Promotion and Development of Engineering and Sciences (SPADES) - Event Head, AI Nexus 2024-2025

- Arranged panel interviews of 300 applicants, selecting 12 recruits systematically.
- Focused on building a diverse team while not sacrificing performance or team dynamics.
- Helped foster the new team's morale via team-building exercises and recreational trips.
- Oversaw department photoshoots, promotional videos, and assigned roles.
- Overlooked the technical development of datasets, models, and competitive aspects, while ensuring proper documentation of the materials.
- Used Trello to assign and manage the progress, even while some members were remote.
- Successfully prepared for 4 months prior to, and conducted a 3-day, 90-participant round, earning the title of the best-managed round across 12 other rounds.